

Promise and Perils of Redeemable Money: Evidence from a Barter Platform

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Abstract

Many digital currencies are backed by promises of redemption. What happens when the promise is broken? This paper answers with transaction-level microdata and three unanticipated policy changes on a Toronto barter platform that operated a redeemable digital token. When the platform restricted and then halted redemption, token acceptance and token-mediated trade fell while the token's posted price stayed at par. The first breach also reduced barter and cut entry of new regular users by two-thirds, as aggrieved users withdrew from the platform. The later halt of all remaining redemption, announced as a pause, provoked no comparable backlash. Before the crises, a five-fold expansion of the token supply had raised trade by 57 percent, entirely through token-mediated exchange, confirming that the token functioned as money. A search-theoretic model of redeemable money with reference-dependent participation organizes these patterns.

Keywords: platform, money, barter, redemption crisis

JEL: E42, E52, E65

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1 Introduction

Digital platforms increasingly issue their own monies. Examples include super-apps such as Ali-pay and WeChat Pay, retailer loyalty currencies, and, more recently, stablecoins. These systems position private firms as currency issuers, trusted to *redeem*: to convert their tokens into goods or official currency on demand. How do redeemability and trust govern whether platform money circulates? What happens when the promise is tested? When a platform currency crisis hits, does the damage stay confined to money-mediated exchange? These questions are central to the stability of emerging monetary systems. Yet the transaction-level evidence needed to study them is scarce.

This paper provides evidence regarding the roles of trust and redeemability in digital currency systems by leveraging a unique micro-dataset from a barter platform that introduced a redeemable currency and then underwent a currency crisis. The setting is Bunz, a community that began in 2013 as a Facebook-based barter network for cash-constrained young adults in Toronto, migrated to a dedicated mobile app in 2016, and banned cash in favor of “true trades” of used goods such as clothing, plants, books, and household items. In 2018 it introduced BTZ, a digital token transferable among users and redeemable at partner “Shop Local” stores at a fixed rate of 100 BTZ to one Canadian dollar. The platform’s records provide high-frequency, transaction-level data on token transfers, item postings, messages, and user ratings, allowing barter and money-mediated trade to be observed together, a combination rarely available.

Three unanticipated changes to the currency provide quasi-experimental variation. The two that are the focus of this paper are reductions in redeemability, each driven by the platform’s finances rather than by economic conditions. In September 2019, facing a cash shortfall, Bunz abruptly restricted redemption to a narrow set of coffee shops and restaurants. In February 2020 it halted redemption entirely. A third, earlier event serves as a complement: between September and October 2018, while redemption was fully operative, Bunz raised token issuance so that the supply rose about five-fold. The broader economy was stable across the sample period, and the app’s release history reveals no confounding design changes.

The analysis makes three contributions. First, it provides, to my knowledge, the first transaction-level field evidence on a platform currency operating alongside barter, spanning the currency’s rise and crisis. Second, it shows that redeemability and trust governed the currency’s viability. A

breach of trust damaged barter as well as token-mediated trade. Third, it characterizes how the currency failed: through acceptance, not price.

To interpret these events, I extend the search-theoretic model of [Kiyotaki and Wright \(1993\)](#). In the canonical model, an intrinsically worthless token facilitates trade by solving the double-coincidence-of-wants problem. I add three features. Tokens are issued and can be redeemed for goods at an exogenous rate. Users choose how much to participate. And their participation depends on the level of redemption they have come to expect.

The first main finding is that redeemability sustained the token's circulation. The model predicts that a lower redemption rate lowers the value of holding tokens and, with it, acceptance and token-mediated trade. Both reductions in redeemability bore this out. After the 2019 restriction, token acceptance, measured as the share of new listings posted with a BTZ price, fell immediately from about 35 to 27 percent platform-wide. Token-mediated trade stabilized about 32 percent below its pre-event level. The 2020 halt reduced both further.

The second main finding is that trust and aggrievement governed the collateral damage. In search models with fixed participation, a reduction in redeemability leaves barter unchanged or raises it. In the extended model, users aggrieved by a breach of the expected redemption level participate less, so barter falls as well. A reduction that users have come to expect adds no aggrievement. The two crises differed along this line. The 2019 restriction reduced barter by nearly a quarter, even though barter requires no token. Entry of new users who went on to become regular fell by two-thirds. Users took the abrupt restriction, which reneged on a pledged notice period, as a betrayal. Affiliated groups dissociated under a "PALZ" label. The 2020 halt removed the token's remaining redemption value. Yet barter returned to near its pre-halt level within months. The halt had been announced as a pause after months of restricted redemption, and it provoked no comparable reaction.

Perhaps surprisingly, posted prices did not detectably adjust in either crisis. The token's median posted price stayed at par, so the adjustment described above fell entirely on acceptance. After the halt that adjustment was slow: acceptance fell at once and then kept eroding for months. The model takes the fixed price as given. A price that stays at a conventional level is a standard possibility in decentralized monetary economies, where the price level is indeterminate (e.g., [Green and Zhou](#),

1998). With backing gone, acceptance in the model rests on expectations of continued acceptance alone. The slow erosion is what results if such expectations adjust gradually rather than at once (Selgin, 2003).

The earlier monetary expansion confirms that the token functioned as a medium of exchange. The five-fold supply increase raised total trades by about 57 percent on a persistent basis. Within existing regular users, trades rose by about 22 percent net of seasonality. The increase is accounted for entirely by token-mediated trade. Barter was unchanged. Token acceptance and the posted price did not change. Velocity fell, indicating that the new tokens relaxed transaction constraints rather than bidding up prices.

The large benefits of money here may owe to Bunz’s cash ban and to its particular goods (used, low-value) and traders (young, urban, transacting with strangers). They need not extend to settings with efficient alternative payments. The broader implications may travel further. Redemption promises plausibly matter for digital-currency circulation elsewhere, and user aggrievement may amplify the costs of other digital-money crises. The contrast between the two episodes suggests that issuers may limit such damage through how they communicate and manage expectations.

Literature. This paper contributes to a large literature on money and barter. Canonical models, dating to Kiyotaki and Wright (1989, 1993), formalize how money arises to overcome the double-coincidence problem (Clower, 1967; Lagos, Rocheteau and Wright, 2017). A growing experimental literature tests these theories in the laboratory (Duffy and Puzzello, 2014; Jiang et al., 2024; Camera, 2024). Direct field evidence from economies where money and barter coexist at high frequency is, however, remarkably scarce: because barter is typically informal and unrecorded, most evidence is historical or cross-sectional (Radford, 1945; Sweeney and Sweeney, 1977; Colacelli and Blackburn, 2009).¹ This scarcity has fueled criticism from historians and anthropologists, who argue that formal models misrepresent real economies and that credit-based accounts better fit the evidence (Humphrey, 1985; Wray, 2004; Graeber, 2011). This paper adds rare high-frequency transaction-level evidence from an economy where money and barter coexisted, and provides real-

¹Radford (1945) described cigarettes as a medium of exchange in a prisoner-of-war camp; Sweeney and Sweeney (1977) recounted how a babysitting co-op on Capitol Hill overcame a recession by expanding its scrip; and Colacelli and Blackburn (2009) used survey responses from the 2002 Argentine crisis to show that participation in fiat-issuing exchange clubs raised consumption among bartering populations.

world validation for this class of models, while documenting margins, especially trust and reputational enforcement, that they typically abstract from (Kehoe and Levine, 1993; Gu et al., 2013a,b; Schnabel and Shin, 2018; Borio, 2019; Gorton and Zhang, 2023).

The paper relates to a rapidly growing literature on digital currencies supplied by private technology platforms. Theory has assigned platform money several roles: as a medium of exchange whose issuance competes with cash and legacy systems (Chiu and Wong, 2022; Ozdenoren, Tian and Yuan, 2025); as a loyalty device that can raise platform profits (Rogoff and You, 2023); as a claim whose price feeds back into platform usage and can fuel speculation (Cong, Li and Wang, 2022); and as a commitment or enforcement device (Brunnermeier and Payne, 2023; Sockin and Xiong, 2023). This paper supplies transaction-level evidence to this literature, supporting the view that platform money helps overcome transaction frictions while underscoring the importance of redeemability and trust for its circulation. A broader literature has studied other novel forms of money, including mobile money (e.g., Jack and Suri 2014), digital payment systems (e.g., Alvarez and Argente 2020a,b; Alvarez et al. 2023), cryptocurrencies (e.g., Alvarez, Argente and Van Patten 2023; Liu, Makarov and Schoar 2023), and central bank digital currency (e.g., Keister and Sanches 2023). Finally, the circulation of BTZ alongside the Canadian dollar relates to classic analyses of currency coexistence, in which the exchange rate between two intrinsically useless currencies is indeterminate (Kareken and Wallace, 1981).

Roadmap. Section 2 describes the Bunz platform, the design of BTZ, and the data. Section 3 presents the search-theoretic framework with redeemable money and reference-dependent participation. Sections 4, 5, and 6 study the monetary expansion, the 2019 restriction, and the 2020 halt. Section 7 compares the two redemption crises. Section 8 concludes.

2 Background

This section describes the Bunz community and the mechanics of trade on its platform, introduces the redeemable BTZ currency, and presents the data and the users who generate them.

2.1 The Bunz Community

Bunz began in 2013 as a Facebook group in which members in Toronto posted items they wanted to give away or acquire. It grew rapidly among cash-constrained young adults, eventually spanning roughly 200 topic- and item-specific groups, some with tens of thousands of members.²

Reflecting the anti-capitalist spirit of its founder, the community had one central rule: no cash. Users were to transact through “true trades”: barter.³ Bunz HQ enforced the rule by removing item postings that requested cash, and the ban was by and large observed: textual analysis of messages between users finds that fewer than five percent of conversations mention cash or dollars.⁴ On the app, users post items (typically a photo, title, description, and location), maintain a profile with an “in search of” (ISO) list, and send offers by browsing and messaging one another. Once a trade is arranged and completed, the two parties rate each other.⁵

The cash ban made the double coincidence of wants a real impediment to trade: interviews indicate that trades frequently failed because the buyer had nothing the seller wanted. When double coincidence was absent, sellers often accepted alternative payments such as beer, gift cards, and transit tokens, typically procured by the buyer just before the trade and used by the seller soon after; users who preferred to barter sometimes declined such offers. Traders were largely strangers transacting bilaterally and simultaneously, with essentially no credit, which lets the analysis focus on money’s role as a medium of exchange.⁶ A public rating-and-review system deterred no-shows and scams, and the resulting high level of trust led many interviewees to prefer Bunz over platforms such as Craigslist and Kijiji, where scams were more common.

²The group was originally named “Bumz.” In early 2016 the community migrated its trading to a dedicated mobile app, built by a separate company (henceforth “Bunz HQ”) with functionality similar to trading apps that later emerged, such as Facebook Marketplace.

³See the Bunz FAQ in Online Appendix B.1.

⁴Interviews with users in May 2019 revealed a range of views on the cash ban. Many, especially Facebook-group administrators, strongly supported it; at least one frequent seller admitted to preferring cash and occasionally steering buyers toward it.

⁵Online Appendix Figure B1 shows the mobile app before and after the introduction of BTZ.

⁶Short-term credit existed but was rare: for example, a deposit to secure a trade, or deferred payment when a new user could not recall a wallet PIN. The general absence of credit and banking is a departure from the macroeconomy.

2.2 The BTZ Token

In April 2018, Bunz HQ introduced “a brand new digital currency,” BTZ, in a major app update, with the stated purpose of facilitating trade. Each user received an endowment of 1,000 BTZ on activating a digital wallet, and could earn more through the “Daily BTZ Drop,” a survey completed by opening the app, and by inviting friends or posting items.⁷ Items could now be posted with a BTZ price, and tokens transferred between users by tapping a profile button or scanning a QR code.

To promote the token and stabilize its value, Bunz HQ created the “Shop Local” program, which let users redeem BTZ for retail goods at partner stores around Toronto, such as coffee shops and restaurants, at a fixed rate of 100 BTZ to 1 Canadian dollar (CAD).⁸ Partner stores could in turn redeem accumulated BTZ for cash from Bunz HQ at the same fixed rate. Apart from the Shop Local program, users could neither buy nor sell BTZ for cash within the app. The total token supply was therefore determined entirely by issuance from the app and redemption by users at partner stores. This one-way, fixed-rate redemption is central to the analysis: it gives the token a fundamental value while leaving its supply under the platform’s control, and the three events studied below are changes to one or the other.

2.3 Data and Who Trades

The data provided by Bunz HQ are rich and comprehensive. I observe the universe of BTZ transfers, with timestamps, amounts, and counterparties, so that every user’s token balance can be reconstructed at any moment; the post-trade ratings, which identify barter trades; all item postings, with descriptions and timestamps; the full text of user-to-user messages; and a range of user characteristics, including geolocation and the Daily BTZ Drop survey responses.

Two data limitations shape the analysis. First, traded items are typically used and non-standardized, and barter trades carry no posted price, so the analysis focuses on the *number* of trades, measured by ratings sent and received, rather than the terms of trade. Second, there is no centralized ex-

⁷The Daily BTZ Drop was designed to raise user traffic and token adoption. The survey collected demographic information used in the descriptive statistics below.

⁸In 2018, the average exchange rate was 1 CAD to 0.77 USD.

Table 1: Regular users versus the broader active-user base

	Regular users	All active users
<i>Demographics (survey respondents)</i>		
Share aged 18–34	69.2%	75.6%
Share with a bachelor’s degree or higher	64.4%	56.0%
Median reported income band	\$35,000–50,000	\$35,000–50,000
<i>Activity (per user)</i>		
Trades, Sept 2018–Aug 2019	30.94	0.68
Items posted	182.28	5.31
Months active	10.1	2.1
Barter share of trades	60.2%	62.1%
Number of users	2,281	215,271

Notes: Regular users are those with at least 50 trades over the full observation period (January 2016 to November 2021), no more than 70 percent of their trades in a single month, and at least six months of activity. “All active users” denotes the full non-Bunz, non-merchant user base. Demographic rows are based on respondents to the voluntary Daily BTZ Drop survey; roughly 60 percent of regular users and five percent of all active users answered the relevant questions. Activity rows are per user and cover September 2018–August 2019. Online Appendix Table B2 reports the distribution of trades among users active for at least six months.

change between BTZ and other currencies; I measure the BTZ price level using the subset of gift cards posted on the platform that carry both a CAD face value and a BTZ price.

Table 1 shows that survey respondents were predominantly young and college-educated, with about seven in ten aged 18–34 and over half holding at least a bachelor’s degree, and spanned a wide range of incomes. Interviews and platform observation suggest the user base also skewed female. Traded goods were highly heterogeneous: clothing and jewelry alone account for about a third of postings, with home goods, groceries, beauty products, electronics, and books also common. The median posted item was priced around 10 CAD (Online Appendix Table B1). This heterogeneity is what makes the double-coincidence problem bite.

A small share of users accounts for most activity.⁹ Much of the analysis therefore focuses

⁹Among all users with at least six months of activity, the top decile accounts for 67 percent of trades and the top five percent for 50 percent, while the bottom two deciles account for essentially none (Online Appendix Table B2).

on *regular users*: those with at least 50 trades over the full observation period (January 2016 to November 2021), no more than 70 percent of their trades in a single month, and at least six months of activity. Because the definition uses activity over the full period, the within-user estimates below describe users who stayed. Between September 2018 and August 2019, regular users were about 1 percent of active users but generated 48 percent of trades, 36 percent of items posted, and 36 to 40 percent of peer-to-peer BTZ transfers sent and received (Online Appendix Table B3). Their use of the platform is highly persistent, with more than 90 percent still active a year after entry (Online Appendix Figure B2).

Because the main estimates focus on regular users, a natural question is how far they differ from the broader user base. Table 1 shows that the differences along observable characteristics are modest: regular users are slightly less likely to be aged 18–34 and somewhat more likely to hold a bachelor’s degree. But they have the same median income band and the same barter share as active users. Users with different levels of trading activity also look similar: barter shares, token acceptance, offers per trade, and BTZ flow sizes vary little across groups (Online Appendix Table B4). Thus the estimates below describe behavior representative of the marketplace in composition, in the goods listed and in location (Online Appendix Tables B5 and B6), estimated on users who differ mainly in engagement and tenure (10 versus 2 months active, Table 1).

3 Conceptual Framework

This section provides a stripped-down extension of [Kiyotaki and Wright \(1993\)](#) to organize the three Bunz events. Relative to the canonical model, it adds three ingredients: redemption and issuance of tokens, an endogenous participation margin, and reference-dependent participation à la [Hart and Moore \(2008\)](#). The framework is deliberately simple and is used as a qualitative benchmark: its value lies in organizing the three events and clarifying which margins each event should move, not in capturing every feature of the data. Proofs and extensions are presented in Appendix A.¹⁰

¹⁰The extended [Kiyotaki and Wright \(1993\)](#) model matches the observed margin structure better than divisible-money models in the tradition of [Lagos and Wright \(2005\)](#). Tokens traded at a par convention (100 BTZ = 1 CAD) that held within a few percent throughout, so adjustment fell on quantities, participation, and acceptance rather than on prices (Section 7). The indivisible-money benchmark builds in this structure and generates the coexistence of

3.1 Environment

Time is continuous and the economy is populated by a unit mass of infinitely-lived agents. Preferences and technology are as in Kiyotaki and Wright (1993): each agent consumes goods produced by others and produces a good of her own; consumption yields utility u , production costs c , with $u > c > 0$; agents discount at rate r . In a meeting, an agent is able to consume the other agent's product with probability $x \in (0, 1)$, and with probability xy the pair has double coincidence of wants, enabling barter. There is no commitment or enforcement beyond immediate trade, so credit is infeasible.

Participation. Agents choose a participation intensity $a \in [0, 1]$ at flow cost $C(a)$, with $C(0) = C'(0) = 0$ and $C'' > 0$. Participation captures the effort of engaging with the platform: listing items, browsing, responding to offers. An agent with intensity a_i meets others at Poisson rate $\alpha a_i \bar{a}$, where $\bar{a}$ is average intensity. For tractability, participation cannot be conditioned on money holdings.¹¹

Tokens, redemption, and failure risk. There is a fixed stock $M \in [0, 1]$ of tokens (“BTZ”), indivisible, with individual holdings in $\{0, 1\}$. At Poisson rate ρ , a token holder has the opportunity to redeem the token at the platform's merchant network, receiving utility u . The platform may also fail: agents attach probability $s \in [0, 1]$ to the redemption commitment being honored. All decisions then depend on (s, ρ) only through *effective redeemability* $\tilde{\rho} \equiv s\rho$: policy changes in ρ and changes in beliefs about platform viability operate through the same sufficient statistic, though they are conceptually distinct events.¹² To keep the token supply constant, a token is issued to a random agent without money at rate $\sigma = \frac{M}{1-M} \tilde{\rho}$, matching the platform's practice of issuance alongside redemption. When $\tilde{\rho} = 0$, the model collapses to the standard Kiyotaki–Wright environment with

equilibria in Proposition 1. In a divisible-goods environment, adjustment operates instead through the token price or redemption flows, and strictly positive backing eliminates the non-monetary equilibrium. The fixed exchange rate is an assumption rather than a prediction (Section 7.2). Zero-or-one holdings omit the intensive margin of balances. A tractable divisible-money model of these episodes is left for future work.

¹¹Participation reflects listing and browsing behavior that is adjusted at lower frequency than individual trades. The assumption keeps the acceptance problem identical across participation levels. Letting participation depend on holdings would allow token holders to search harder than non-holders, so that token-mediated trade could rise without barter; the benchmark keeps the two margins tied.

¹²This is the sense in which platform failure is observationally equivalent, within the model, to a deterioration of redemption policy. The empirical sections use announcement timing to separate the two.

intrinsically worthless fiat money.

Reference-dependent participation. Agents hold a reference level of redeemability $\tilde{\rho}_0$ equal to the redemption they expect. Following [Hart and Moore \(2008\)](#), when realized redeemability falls short of this reference ($\tilde{\rho} < \tilde{\rho}_0$), agents are aggrieved and shade their participation: the participation cost carries an additional flow term $\lambda a G$, with $\lambda \geq 0$, where G is the largest shortfall $(\tilde{\rho}_0 - \tilde{\rho})^+$ the agent has experienced to date. Two features matter for the analysis below. First, aggrievement responds to the *unanticipated* component of a reduction: a fall below the prevailing expectation $\tilde{\rho}_0$ aggrieves users, whereas one they have already come to expect, so that $\tilde{\rho}_0$ has adjusted to the new level, does not. Second, grievance persists once incurred: a downward revision of expectations does not undo a breach already suffered. The contrast between the two reductions studied below therefore turns on the reference $\tilde{\rho}_0$, that is, on whether each was anticipated, rather than on its size.

3.2 Equilibrium and the Role of Redeemability

I focus on stationary symmetric equilibria, in which all agents with the same money holdings behave identically ([Wright, 1999](#)). Let π denote the probability that a trade in a single-coincidence meeting is consummated with a token, and let V_1 and V_0 denote the values of holding one and zero tokens. The key term in the Bellman equation for V_1 (Appendix A) is the expected gain from redemption, $\tilde{\rho}(u + V_0 - V_1)$, so the seller's gain from accepting a token turns positive as $\tilde{\rho}$ rises: redeemability gives the token a fundamental value. A seller's willingness to produce in exchange for a token therefore rises with both $\tilde{\rho}$ and the acceptance of others, the self-fulfilling acceptability logic of [Kiyotaki and Wright \(1992\)](#). Following Section 3 of [Rupert et al. \(2000\)](#), agents do not use tokens in double-coincidence meetings, so barter and monetary trade are substitute forms of exchange.

Proposition 1 (A pure-strategy benchmark). *Assume $M < \bar{M} \equiv 1 - c/u$ and fix participation $a > 0$. There exist thresholds $\underline{\rho} < \bar{\rho}$, with $\bar{\rho} > 0$, such that in pure strategies:*

1. *For $\tilde{\rho} > \bar{\rho}$, only monetary equilibria ($\pi = 1$) exist.*
2. *For $\tilde{\rho} < \underline{\rho}$, only non-monetary equilibria ($\pi = 0$) exist.*
3. *For $\tilde{\rho} \in [\underline{\rho}, \bar{\rho}]$, monetary and non-monetary equilibria coexist.*

Proof. Appendix A. □

Proposition 1 captures the idea that reliable redemption can make tokens widely acceptable, while insufficient redemption leaves the economy exposed to a self-fulfilling collapse of acceptance: in the coexistence region, whether tokens circulate depends on beliefs. Two cautions about what the proposition does and does not say. First, it characterizes pure-strategy outcomes only. The unique mixed-strategy equilibrium (Lemma A.1 in Appendix A) has counter-intuitive comparative statics, since acceptance must *rise* as $\tilde{p}$ falls to maintain indifference, but it is the unstable tipping point separating the basins of attraction of the two pure equilibria, not a state in which the economy can persist. Second, the benchmark cannot generate persistent *partial* acceptance, which the data display. Appendix A.5 develops an extension with heterogeneous redemption access à la Shevchenko and Wright (2004), in which stable equilibria feature interior aggregate acceptance that declines smoothly as redeemability falls; the pure-strategy benchmark should be read as the homogeneous limit of that environment.

3.3 Participation and the Backlash Channel

In a symmetric stationary equilibrium with acceptance π , the expected total surplus per meeting is

$$\bar{S}(\pi) = (u - c) \left[\underbrace{xy}_{\text{barter}} + \underbrace{x(1-y)M(1-M)\pi}_{\text{monetary trade}} \right], \quad (1)$$

and the participation intensity a^* satisfies the first-order condition (Appendix A):

$$\underbrace{\alpha a^* \bar{S}(\pi)}_{\text{marginal surplus}} = \underbrace{C'(a^*)}_{\text{effort cost}} + \underbrace{\lambda G}_{\text{aggrievement}}. \quad (2)$$

Trade flows are $T_B = \frac{1}{2}\alpha a^{*2}xy$ for barter and $T_M = \alpha a^{*2}x(1-y)(1-M)M\pi$ for token-mediated trade.¹³

¹³Because matching is $\alpha a_i \bar{a}$, $a = 0$ is always an equilibrium: participation is worthless when no one else participates. I select the largest solution of (2), which exists and is interior when $C''(0) < \alpha \bar{S} < C'(1)$ and $\lambda G < \max_a [\alpha a \bar{S} - C'(a)]$; there $C''(a^*) \geq \alpha \bar{S}$, generically strictly, so it is locally stable, and all comparative statics are evaluated at it.

Two features of (1)–(2) drive the predictions. First, the barter surplus xy depends on neither $\tilde{\rho}$ nor M , so both reach barter T_B only through participation a^* . Second, a fall in $\tilde{\rho}$ below the reference, which raises G one-for-one, lowers participation through two channels:

$$\left. \frac{da^*}{d\tilde{\rho}} \right|_{\tilde{\rho} < \tilde{\rho}_0} = \underbrace{\frac{\alpha a^* \partial \bar{S} / \partial \tilde{\rho}}{C''(a^*) - \alpha \bar{S}}}_{\text{token value}} + \underbrace{\frac{\lambda}{C''(a^*) - \alpha \bar{S}}}_{\text{aggrievement}} \geq 0. \quad (3)$$

Here the first term is the token-value channel, which operates only through acceptance, since $\bar{S}$ depends on $\tilde{\rho}$ solely via π (smoothly in the extension of Appendix A.5, by a jump between equilibria in the benchmark); the second is aggrievement. Both lower barter through participation, so a barter decline alone does not reveal which is responsible.¹⁴

Issuer discipline. The same aggrievement makes the redemption promise credible. Suppose the platform earns flow value ϕ_P per trade and bears cost κ_R per unit of redemption flow. Maintaining redeemability is then self-enforcing when the trade value a breach destroys exceeds the liability it saves:

$$\phi_P [T(a^*(\tilde{\rho}_0), \pi(\tilde{\rho}_0)) - T(a^*(\tilde{\rho}'), \pi(\tilde{\rho}'))] \geq \kappa_R M (\tilde{\rho}_0 - \tilde{\rho}'), \quad (4)$$

with $T = T_B + T_M$ total trade and $\tilde{\rho}' < \tilde{\rho}_0$ the breach. The left side rises in λ : a stronger backlash means a larger trade loss from a breach, widening the range over which the promise is kept. The same parameter that drives the post-breach barter decline thus disciplines the issuer beforehand, and a breach occurs when the redemption liability outgrows the platform's valuation of future activity.¹⁵

3.4 Empirical Predictions

The three Bunz events map into the framework as follows. Throughout, the economy starts in a monetary equilibrium ($\pi = 1$, $\tilde{\rho} = \tilde{\rho}_0 > \bar{\rho}$) with $M < \min\{\bar{M}, \frac{1}{2}\}$ (Appendix A); for Prediction 1,

¹⁴The participation margin nests exit. With heterogeneous effort costs, a large enough aggrievement wedge pushes some users to $a = 0$ and deters entry. The backlash then operates on both the intensive and extensive margins.

¹⁵Equation (4) is a deliberately simple comparison of stationary flows rather than a full repeated game; relational-contracting or limited-commitment microfoundations à la Kehoe and Levine (1993) would endogenize the same trade-off, with Hart and Moore (2008) supplying the microfoundation for the aggrievement response itself.

$\tilde{\rho} > \bar{\rho}$ is assumed to hold at the post-expansion stock as well, since $\bar{\rho}$ rises with M . All predictions compare stationary equilibria, so the changes are persistent.

Prediction 1 (Monetary expansion). *Suppose M increases, with $\tilde{\rho}$ fixed. Then:*

- 1.1 T_M and total trade $T = T_B + T_M$ increase;
- 1.2 Acceptance π is unchanged, and G stays at zero;
- 1.3 Participation a^* (weakly) increases;
- 1.4 T_B moves only through a^* , whereas M raises T_M directly.¹⁶

Prediction 2 (Unanticipated reduction). *Starting from $\tilde{\rho} = \tilde{\rho}_0$, an unanticipated reduction lowers $\tilde{\rho}$ to $\tilde{\rho}' < \tilde{\rho}_0$. Then:*

- 2.1 Acceptance π , T_M , and T fall;¹⁷
- 2.2 Users are aggrieved: G rises to $\tilde{\rho}_0 - \tilde{\rho}'$;
- 2.3 Participation a^* falls through both terms of (3), so T_B falls by more than the token-value term alone.

Prediction 3 (Anticipated reduction to zero). *The reference has adjusted to $\tilde{\rho}_0 = 0$, and the halt takes $\tilde{\rho}$ to 0. Then:*

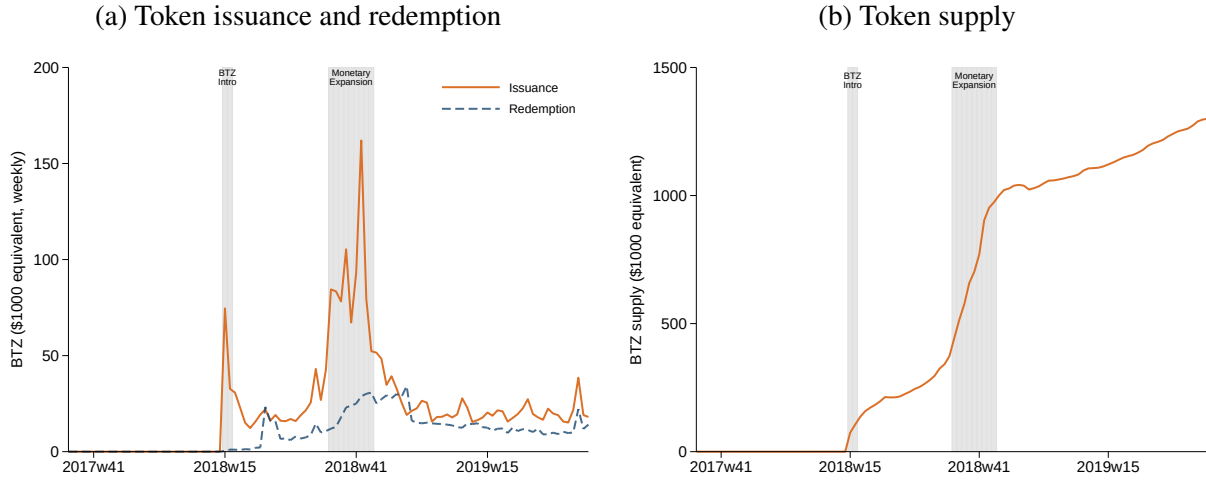
- 3.1 Acceptance π and T_M fall further;
- 3.2 There is no aggrievement: G is unchanged;
- 3.3 Participation a^* falls through the token-value term of (3) alone, so T_B falls only insofar as the token's value sustained participation.

Section 4 tests Prediction 1 with the September–October 2018 expansion, Section 5 tests Prediction 2 with the September 2019 restriction, which was abrupt and unforeseen, and Section 6 tests Prediction 3 with the February 2020 halt.

¹⁶From (1), M raises the return to participation only via the term $M(1 - M)\pi$, so the barter response is small when participation is inelastic.

¹⁷In the benchmark, a fall in $\tilde{\rho}$ below $\underline{\rho}$ eliminates the monetary equilibrium; a fall into the coexistence region leaves acceptance to beliefs but raises the tipping point of Lemma A.1, so acceptance unravels more easily. The persistent partial decline is the comparative static in Appendix A.5, where aggregate acceptance falls monotonically in $\tilde{\rho}$ (Proposition A.1).

Figure 1: Token issuance, redemption, and supply, before and after monetary expansion



Notes: Panel (a) shows the weekly trend in BTZ issuance, the total amount of tokens sent from Bunz directly to users, and BTZ redemption, the total amount of tokens sent from users to local stores. Panel (b) shows BTZ supply, the cumulative sum of BTZ issued minus the cumulative sum of BTZ redeemed. All numbers are denominated in the Canadian dollar (CAD) at the fixed exchange rate: 1 CAD = 100 BTZ. Gray bars indicate the first and second wave of monetary expansion.

4 Effects of Monetary Expansion

This section empirically tests Predictions 1.1–1.4 by estimating the effects of a large monetary expansion on trade on the Bunz platform. Consistent with the predictions, it is found that monetary expansion increased token-mediated trade, but did not detectably affect barter trade.

4.1 Timeline of Events

Between September and October 2018, roughly half a year after the initial introduction of the token, Bunz HQ increased the Daily BTZ Drop to 100 BTZ per day from 10 BTZ per day, in hopes of increasing use of the token.¹⁸ To the best of my knowledge, this increase was unanticipated by users. After roughly seven weeks of increased token issuance, Bunz HQ realized that the resulting pace of token redemption would be financially unsustainable, and reverted to Daily BTZ Drops of 10 BTZ per day.

Figure 1 Panel (a) plots weekly issuance (orange) and redemption (blue). Issuance came in two

¹⁸The Daily BTZ Drop functions as a “helicopter drop” of money (Friedman, 1969; Bernanke, 2002): tokens are credited directly to users’ accounts.

waves. A small wave occurred in April 2018 (Week 15), when each user activating a BTZ wallet received 1,000 BTZ (about 10 CAD). The large wave, the focus of this section, came roughly five months later, in Weeks 36–42. Redemption rose after each wave, briefly in May 2018 and again from October to December 2018, but fell back toward its initial level by the end of the year. Issuance therefore greatly exceeded redemption.

Figure 1 Panel (b) shows the resulting token supply, cumulative issuance net of redemption. Supply stabilized near 20 million BTZ (0.2 million CAD) after the April–May wave. It then grew rapidly from September 2018 and settled after November at roughly 100 million BTZ (1 million CAD). The supply thus rose about five-fold and stayed there.

4.2 Empirical Strategy

How did monetary expansion affect trade? I measure trade volume by the ratings that users give each other. Trades occurred offline and were not recorded directly, so ratings are the best available proxy (Section 2). They undercount trades, since not every trade was rated. But they are available before and after the introduction of BTZ, whereas token transfers are observed only after it. I classify a trade as token-mediated if a token transfer occurred between the same pair of users within 7 days of the rating, and as barter otherwise.

To study the impact of the monetary expansion, I focus on data from two years prior to the monetary expansion to one year after, thereby excluding the impacts of subsequent events.

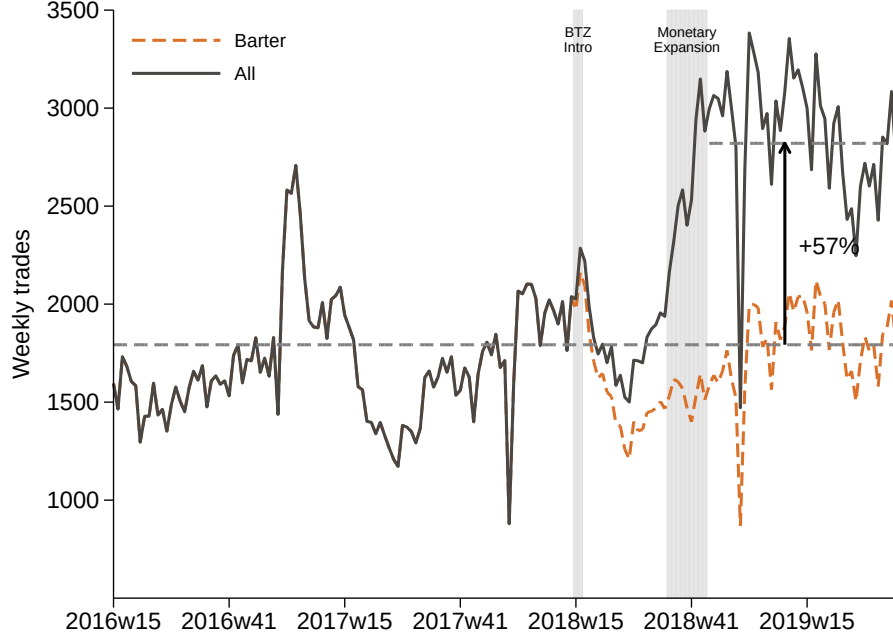
First, I plot the trends in total trades among all users. The changes in levels are computed by comparing averages during the two years prior to monetary expansion to those during the year after. This approach allows me to visually inspect the aggregate change at the platform level.

Second, I analyze the effects of monetary events at the individual account level. Specifically, I construct a monthly panel of regular users, excluding users who entered the platform after the expansion (as measured by first messages sent). I then regress:

$$y_{it} = \beta PostExpansion_t + \delta_i + \delta_{c(t)} + f(t - \tau(i)) + \varepsilon_{it}, \quad (5)$$

where y_{it} indicates user-level outcomes, $PostExpansion_t$ indicates that month t is after September

Figure 2: Effect of monetary expansion on trade volume, all users



Notes: Figure shows the weekly trend in the number of peer-to-peer trades, as measured by user reviews. Barter trades are the subset of trades that did not coincide with a token transfer between the same user pair within 7 days. Gray bars indicate the first and second wave of monetary expansion.

2018, δ_i are user fixed effects, $\delta_{c(t)}$ are calendar month fixed effects, $f(t - \tau(i))$ indicates controls for time since user entry, and ε_{it} is an error term. Standard errors are clustered at the user level.

Several threats to identification are addressed by this specification. User composition is held fixed by restricting to regular users who entered before the expansion and including user fixed effects; seasonality is absorbed by calendar-month fixed effects; and the natural decline in activity with tenure is controlled by the $f(t - \tau(i))$ term in Equation (5). Under these controls, β recovers the causal effect provided existing users' outcomes would otherwise have tracked those of earlier entry cohorts. Two further confounds are unlikely to matter: aggregate consumption in Ontario was flat over the period (Online Appendix Figure B3) and the app's release history shows no contemporaneous design change.

4.3 Main Results

Total trade persistently increased. The black line in Figure 2 plots the trend in total trades. In the two years before the introduction of BTZ, total peer-to-peer trades were largely stable. After the monetary expansion in September 2018, total trades persistently increased by 57 percent, and remained significantly elevated for a year. This finding is consistent with Prediction 1.1.

Barter trade did not change. The orange line in Figure 2 shows that monetary expansion did not change the overall level of barter trades (i.e., trades not associated with a contemporaneous token transfer) among all users. The rise in total trade volume is instead accounted for by the emergence of token-mediated trades. These token-mediated trades began to emerge almost immediately following BTZ introduction, but substantially grew during the monetary expansion. This finding is consistent with Prediction 1.4.

Observed impacts are not due to shifts in user composition. Table 2 reports regression results for existing regular users, with controls for user fixed effects and time since user entry. Column (1) shows that the effect of monetary expansion on total trades among existing regular users is positive and statistically significant, at 0.30 in asinh units, about 35 percent. Column (2) adds controls for seasonality, and shows that the increase is attenuated but still significant, at 0.20, about 22 percent. Columns (3) and (4) show that barter trades among existing regular users did not significantly change, even after controlling for user fixed effects, time since entry, and seasonality. The increase in total trade is similar for users with different levels of activity, and barter rises only in the lowest-activity group, by 0.05 in asinh units, in a specification without calendar-month controls (Online Appendix Table C1). The rate of entry by regular users onto the Bunz platform did not change during the monetary expansion (Online Appendix Figure B4).

4.4 Mechanisms

What explains the increase in trade? The framework offers two channels. Added money raises token-mediated trade directly, because more meetings involve a token holder. It also raises participation, because trading becomes more rewarding. The two channels differ in what they do to barter. If we assume, as the framework above does, that tokens are used only when wants do

Table 2: Effects of monetary expansion on trade volume, existing regular users

	(1) Asinh total trades	(2) Asinh total trades	(3) Asinh barter trades	(4) Asinh barter trades
Post expansion	0.296** (0.023)	0.199** (0.028)	0.032 (0.022)	-0.044 (0.027)
User FE	X	X	X	X
Seasonality		X		X
Observations	49942	49942	49942	49942
Users	1815	1815	1815	1815
R^2	0.293	0.299	0.303	0.308
Pre-event mean	1.232	1.232	1.207	1.207

Notes: Table reports the effects of monetary expansion on asinh total trades and asinh barter trades of existing regular users. Sample includes all observations between October 2016 and September 2019 at the month-user level for regular users whose first message sent was before the monetary expansion. “Post expansion” is defined as months after September 2018. Controls for user fixed effects and months after user entry are included. Controls for calendar month fixed effects are added in Columns (2) and (4). Standard errors are clustered at user level. * and ** indicate statistical significance at the 10 and 5 percent level, respectively. $\text{asinh}(x) = \ln(x + \sqrt{x^2 + 1})$ is used to account for zeros.

not coincide, then the direct channel leaves barter unchanged. The participation channel instead raises barter and token-mediated trade alike. Since it is observed that barter did not change, the evidence suggests that participation responded little. Token acceptance did not change either, as the framework predicts when redemption is intact.

The rise in trade decomposes into offer messages sent and trades completed per offer. Offer messages jumped 41 percent immediately after the expansion, while trades per offer rose 11 percent (Online Appendix Figure C1). Among existing regular users, offers rose by 0.25 to 0.39 in asinh units (28 to 48 percent) net of user fixed effects and tenure, the lower figure also net of seasonality (Online Appendix Table C2). Buyers could see which goods a seller accepted before making an offer. The surge in offers is therefore consistent with both channels. Tokens let buyers make offers on items they could not barter for, which is the direct channel. More offers per user also indicate greater participation (Prediction 1.3). The supply of goods responded as well, though more gradually. New items posted rose with a lag (Online Appendix Figure C2 and Table C2).

Token acceptance, measured by the share of items posted with a BTZ price, rose steadily with the growing number of redemption stores but did not jump at the expansion, stabilizing near 35 percent (Online Appendix Figure C3). A formal test finds no change in the acceptance trend at the end of the expansion once the number of active redemption merchants is controlled for, and only a marginal level shift (Online Appendix Table C3), consistent with Prediction 1.2.

Velocity fell about two months after the expansion, and a break imposed in the final week of 2018 is large and significant (Online Appendix Figure C4 and Table C4). The decline is the accounting counterpart of rigid prices: supply rose faster than trade while the posted token price stayed within a few percent of par (Section 7.2), so each token changed hands less often.

5 Effects of Reduced Redeemability

This section estimates the effects of the 2019 restriction on transaction behavior on the Bunz platform. The estimates confirm Prediction 2.1, which states that reduced redeemability should cause token acceptance and token-mediated trade to persistently decline. Barter also declined, partly through reduced user entry. The decline exceeds what the token's reduced value alone can explain. The excess is the aggrievement effect of Prediction 2.3: users aggrieved by the breach participated less.

5.1 Timeline of Events

The reduction in redeemability was prompted by cash-flow difficulties. As 2019 progressed, Bunz HQ's financial position became untenable: with roughly 18 employees on its payroll, token redemption draining its coffers, and advertising sales insufficient to offset the outflow, neither new investment nor an acquisition was forthcoming (Galang, 2019).

In the weeks *before* the announcement, a small number of users exploited sign-up and referral bonuses (1,000 and 500 BTZ) to create numerous accounts and immediately redeem the tokens at a single store; these are the pre-event spikes in Online Appendix Figure D1 Panel (a). One user redeemed 304,778 BTZ at once. According to Bunz's CEO, this activity contributed to the decision to reduce redeemability in order to forestall a larger run.

On September 9, 2019, Bunz HQ announced that tokens would henceforth only be redeemable at local partner stores selling coffee or food. In a letter sent to Shop Local partner businesses, Bunz HQ wrote, “Effective immediately, you will no longer be able to accept BTZ and convert them into CAD currency. We will be locking your wallets, and everyone will be paid up to September 10th inclusive” (Online Appendix D.1).

Shocked and disgruntled, Shop Local partners announced the changes to the wider Bunz community on the app. They criticized Bunz HQ for the abruptness of the decision, its lack of transparency, and its reneging on a promise to give 30 days’ notice of changes to the Shop Local program. One wrote, “While I respect their decision to end the program, more notice would have been nice. This was literally [zero] notice and not professional. I now have customers that can no longer support me on this platform, many who saved BTZ for months. And now their BTZ is no longer of use to them.”

The next day, Bunz HQ provided an update to the broader community in a blog post. The post confirmed that it would no longer accept BTZ except at coffee shops and restaurants. Apologizing for “any inconvenience and disappointment this may have caused”, the blog post went on to explain that Bunz HQ also had to make the difficult decision to lay off 15 employees that same day.¹⁹ Users reported a tremendous uncertainty about the token’s viability in the following days.²⁰

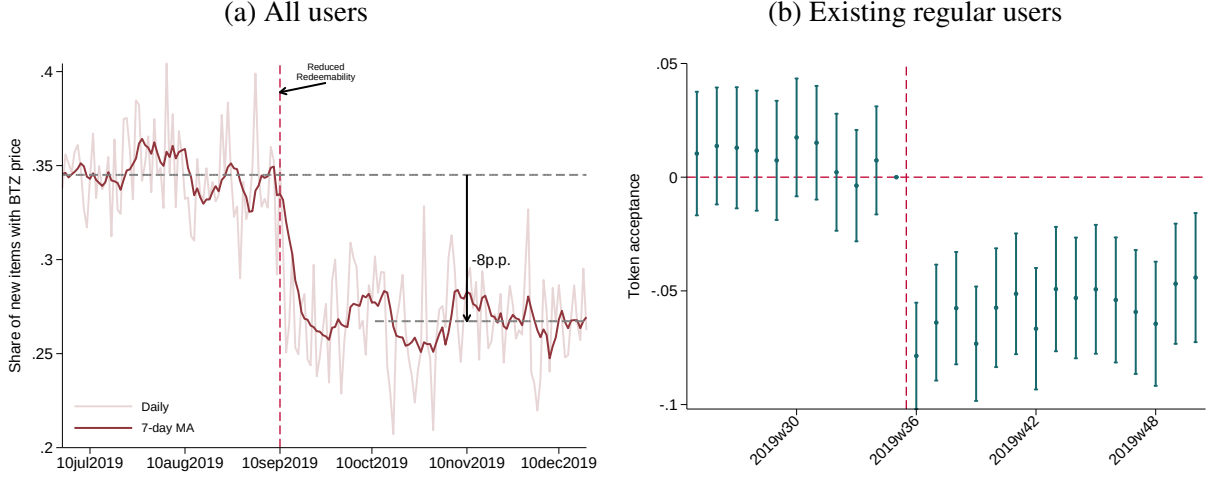
5.2 Empirical Strategy

To estimate the impact of reduced redeemability, I focus on data from 11 weeks before the reduction until 15 weeks after. As before, aggregate trends are first examined. Then, I estimate the effects of reduced redeemability on existing regular users using an event study design at the user level. I construct a weekly panel of regular users, but exclude users who enter after reduced

¹⁹Online Appendix D.2 provides the full text. The digital wallets of employees were also suddenly locked (Galang 2019).

²⁰In an interview, a frequent seller of used books recounted stopping accepting BTZ after the announcement and spending down his balance at restaurants, “eating like a king.” Two weeks later he found that BTZ traded among users at a discount but remained redeemable at restaurants, so he resumed accepting it, at 85 to 90 percent of face value as of October 18, 2019. He redeemed immediately, keeping only a small balance to limit his exposure should the token become worthless (Online Appendix G).

Figure 3: Effects of reduced redeemability on token acceptance



Notes: Panel (a) shows the trends in the share of new items with a posted BTZ price. The dark lines show the 7-day moving average, while the light lines show the daily trend. The red dashed line indicates September 10, the day redemption was restricted. Panel (b) reports individual-level regression estimates, showing that among existing regular users, token acceptance fell by about 7 percentage points.

redeemability, and regress:

$$y_{it} = \sum_{k=-11}^{15} \beta_k I_{t-t^*=k} + \delta_i + \varepsilon_{it}, \quad (6)$$

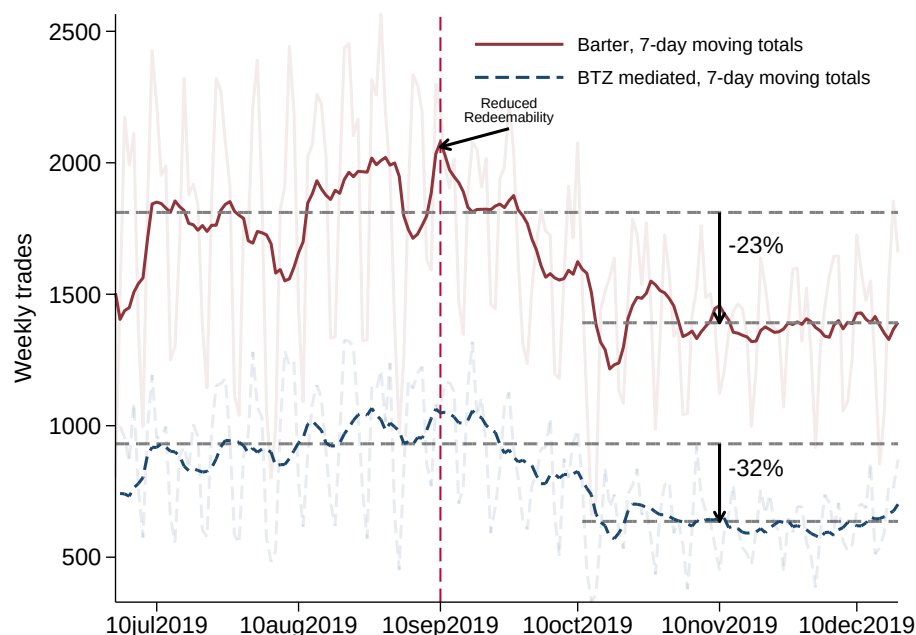
where t^* is the event week, y_{it} indicates user-level outcomes, $I_{t-t^*=k}$ are dummies indicating each week relative to when redeemability was reduced, δ_i are user fixed effects, and ε_{it} is an error term. Standard errors are clustered at the user level.

The coefficients β_k recover the causal effect under the assumption that outcomes would otherwise have tracked earlier entry cohorts, which is assessed from the pre-event coefficients. As in Section 4, stable Ontario consumption (Online Appendix Figure B3) and the app's unchanged release history make confounding by broader shocks or platform design unlikely.

5.3 Main Results

Token acceptance immediately fell. Figure 3 shows the impact of reduced redeemability on token acceptance. Panel (a) shows that during the two months before the announcement, token acceptance, measured as the share of new items with a posted BTZ price, hovered around 35 percent. After the announcement, the share immediately plunged to roughly 27 percent. Panel (b)

Figure 4: Effects of reduced redeemability on trade volumes, all users



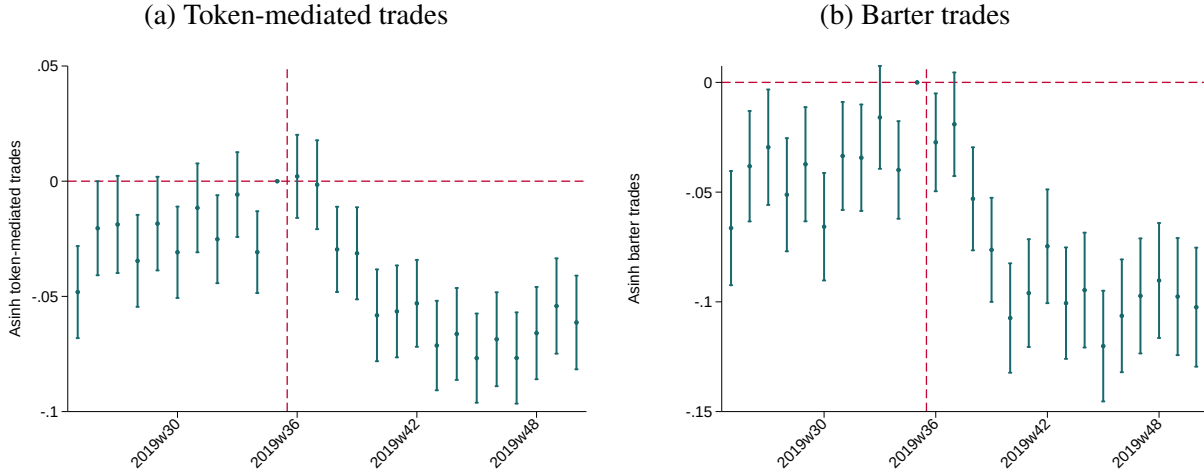
Notes: Figure shows the trend in the number of transactions decomposed by whether a token transfer occurred between the same user pair within 7 days. The dark lines show the 7-day moving average, while the light lines show the daily trend. The red dashed line indicates September 10, the day redemption was restricted.

reports regression coefficients for Equation (6), with share of items with BTZ price at the individual level as the outcome. Token acceptance immediately fell by around 7 p.p. among existing regular users, conditional on their posting new items, even after controlling for user fixed effects and time since entry. The decline is unconfounded by the presence of pre-event trends.²¹ This finding is consistent with Prediction 2.1.

Token-mediated trade persistently fell. The dashed blue line in Figure 4 shows that before redeemability was reduced, the number of token-mediated peer-to-peer transactions, as measured by user ratings associated with a concurrent token transfer, was largely stable. However, after the announcement, it began to decline gradually, stabilizing at 32 percent lower, roughly one month

²¹Online Appendix D.3 provides documentary evidence of reluctance to accept the token from the Bunz platform. One user wrote that “I don’t accept BTZ anymore due to uncertainty. I believe BTZ & BUNZ will cease to exist shortly.” Another wrote, “I’m paused on BTZ for now, until we get some stability.” Yet another wrote, “I will only be doing TRUE TRADES from now on. I no longer believe that BTZ is a sustainable form of currency because of the lack of choices that the users have, and the fluctuating rate at which they are rewarded.”

Figure 5: Effects of reduced redeemability on trade volumes, existing regular users



Notes: Figure reports coefficients from regressions of (a) asinh token-mediated trades and (b) asinh barter trades on week fixed effects, with controls for user fixed effects and months after user entry. The sample includes only existing regular users. Standard errors are clustered at user level. 95% confidence intervals are shown.

after the announcement. This finding is consistent with Prediction 2.1.

Barter trade persistently fell. The solid red line in Figure 4 shows that before the reduction in redeemability, barter transaction volume was largely stable. Immediately after the announcement, however, it began to fall, with a trend break around the date of the announcement, eventually stabilizing at 23 percent lower roughly a month later. This decline is larger than a no-backlash benchmark ($\lambda = 0$) delivers, under which barter responds to redeemability only through the token's diminished value, and it is the evidence for the backlash channel in Prediction 2.3.

Declines occur within existing users as well as through entry. The aggregate reduction in trade and platform usage is partly attributable to reduced entry: the number of new users per month who went on to become regular users fell by roughly two-thirds after the restriction, relative to the year before (Online Appendix Figure B4). The rest reflects reduced activity by continuing users. Panel (a) of Figure 5 reports individual-level regression estimates, showing that among existing regular users, token-mediated trade fell by about 0.05 in asinh units, roughly a quarter of the pre-event mean of the transformed outcome (0.18). The decline is persistent and unconfounded by the

presence of pre-event trends. Panel (b) of Figure 5 shows the corresponding estimates for barter trade, revealing an average longer-run decline of about 0.08 in asinh units, roughly a quarter of the pre-event mean (0.31), after controlling for user fixed effects and time since entry. Declines occur in every activity subsample and are largest among the heaviest traders (Online Appendix Table D1).

5.4 Effects Beyond Token Trade

The online appendix shows that confidence in the token was visibly shaken. Realized redemptions spiked immediately after the announcement and stayed elevated for about two weeks before settling roughly 30 percent below their pre-event level. Among existing regular users the redemption coefficient is 0.38 in asinh units in the first four weeks, more than the pre-event mean of 0.31, and 0.16 thereafter (Online Appendix Figure D1). The token supply, however, barely contracted, from a peak of 130.5 to 127.5 million BTZ (Online Appendix Figure D2), because restricting redemption to small, perishable purchases at coffee shops and restaurants made it highly frictional.

The crisis was not confined to token trade, however. New items posted fell by roughly 29 percent (Online Appendix Figure D3). Offer messages fell by roughly 15 percent, after a brief spike as users spent down balances (Online Appendix Figure D4). Entry and within-user trade fell as well (Section 5.3). Together, these declines describe a withdrawal of participation, the extensive margin of a in the framework, rather than a contraction confined to token acceptance.

6 Effects of Redemption Halt

This section tests Prediction 3 by estimating the effects of the final halt of redemption by Bunz HQ on trade. Consistent with theory, I find an immediate reduction in token acceptance and a more gradual reduction in token-mediated trade volume. After the Covid lockdown, aggregate barter fully recovered, so that, unlike after the 2019 restriction, the lasting damage was largely limited to token-mediated trade.

6.1 Timeline of Events

The final halt was again driven by cash-flow difficulties: even the scaled-down Shop Local program continued to drain tokens. By this point almost all employees had been laid off and the CEO had departed; management passed to an investor who ran the app on advertising revenue, with only the community manager remaining.

The pause to the Shop Local program was announced on February 28, 2020. Unlike after the 2019 restriction, the platform said that the complete halt in redemption was due to technical difficulties and would be “temporary.” During this time, the platform would address “the gaming of BTZ rewards” by “implementing additional checks and controls, which [they] will communicate out once those controls are in place” (Online Appendix [E.1](#)). The Shop Local program was never restarted thereafter.

After the redemption halt, token redemption immediately and persistently dropped to zero. Token issuance also dropped immediately, indicating that users were much less likely to answer the “Daily BTZ Drop” surveys or take other actions on the platform that rewarded users with tokens.

6.2 Empirical Strategy

Measurement of the long-run impact is complicated by the arrival of the Covid-19 pandemic about two weeks after the halt. Toronto transit usage was at normal levels until March 12, 2020, then fell sharply as schools closed; Ontario declared a state of emergency on March 17, closing most businesses, and began relaxing restrictions only in mid-May ([TTC Board, 2020](#); [Nielsen, 2020](#)).

Because of the above, the effect of the redemption halt cannot easily be separated from the effects of the Covid-19 pandemic except in the first two weeks. Following previous sections, aggregate trends for all users are first examined. Then, I estimate the effects of the redemption halt on existing regular users using an event study design. Specifically, I construct a weekly panel of regular users, spanning the 20 weeks before the event until 34 weeks after, but excluding users

who enter after the halt, and regress:

$$y_{it} = \sum_{k=-20}^{34} \beta_k I_{t-t^*=k} + \delta_i + \varepsilon_{it}, \quad (7)$$

where t^* is the event week, y_{it} indicates user-level outcomes, $I_{t-t^*=k}$ are dummies for each week relative to the halt, δ_i are user fixed effects, and ε_{it} is an error term. Standard errors are clustered at the user level. The roughly two-week gap between the halt and the lockdown provides a clean window for the immediate effects, most importantly the fall in token acceptance, which occurs entirely before Covid. The long-run estimates are contaminated by the pandemic and are descriptive rather than causal. The contrast between barter and token trade rests on their differential recovery, on the assumption that the lockdown depressed both margins alike.

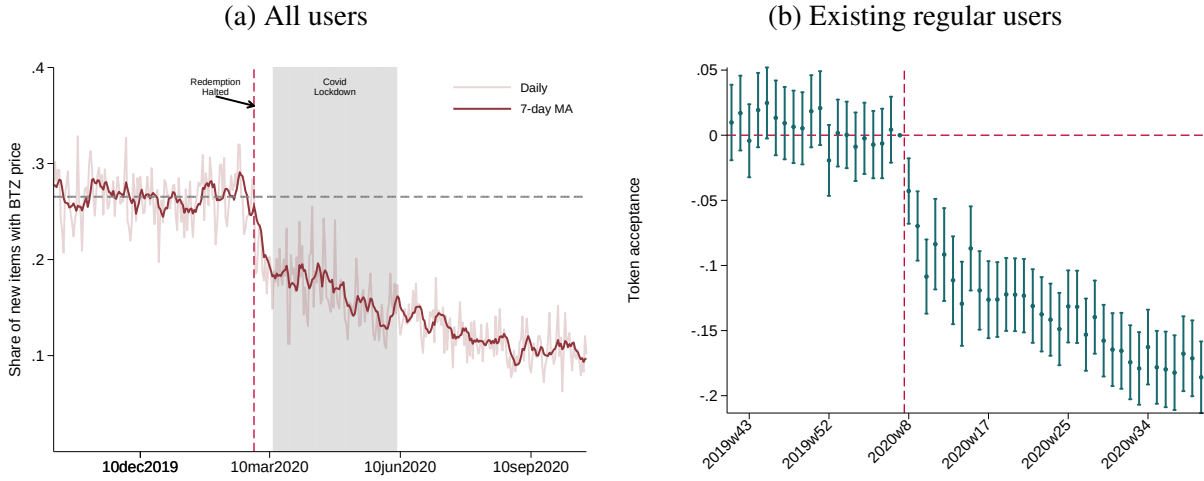
6.3 Main Results

Token acceptance immediately fell. Figure 6 Panel (a) shows that token acceptance immediately fell after redeemability was halted. The share of new items with a posted BTZ price hovered around 26 percent during the two months before the halt. After the halt, the share immediately plunged to roughly 19 percent. This reduction occurred *before* the onset of Covid-19, so it is attributable to the redemption halt. Thereafter, the share continued to slowly slide downward without any interruption from Covid-19. By the end of 2020, only about 10 percent of item posts had a BTZ price.²²

Panel (b), estimated from Equation (7), shows that token acceptance among existing regular users fell immediately, by 4 percentage points in the first week after the halt and 7 in the second, both before the lockdown, and by about 10 points after the lockdown ended, conditional on posting items, even after controlling for user fixed effects and time since entry. There was not only a complete absence of pre-event trends and a sharp drop in token acceptance, but token acceptance among existing users did not recover, and the decline then deepened over time. Estimates across activity-intensity subsamples are noisier but point the same way (Online Appendix Table E1). The

²²As of June 2021, it was exceedingly rare for items to still be posted with a BTZ price. Once while using the app, I encountered a user who was willing to accept BTZ in exchange for a used book. But upon further inquiry, I learned that this was because he had hoped to give the item to someone who could use it, since “BTZ right now has \$0 value.”

Figure 6: Effects of redemption halt on token acceptance



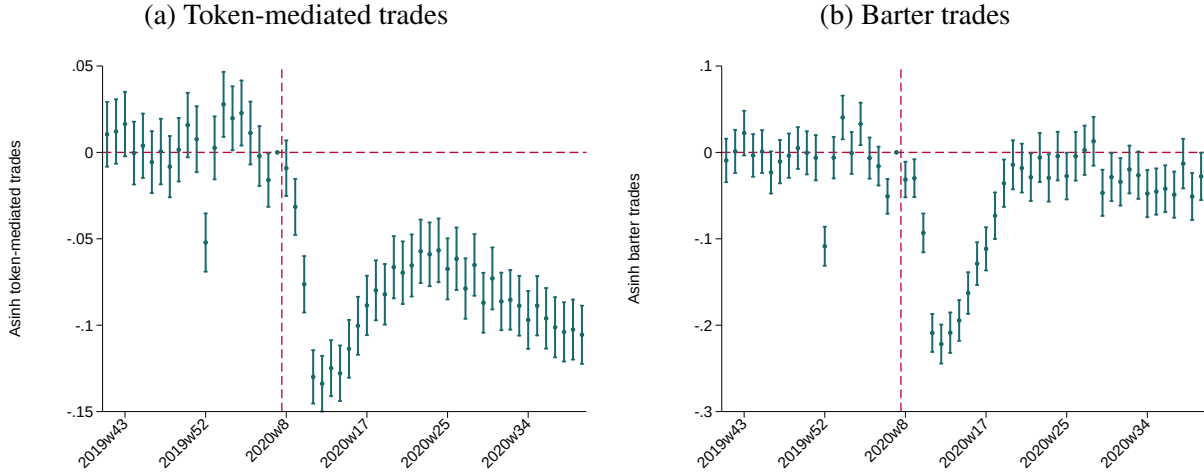
Notes: Panel (a) shows the share of new items with a posted BTZ price. The dark lines show the 7-day moving average, while the light lines show the daily trend. The red dashed line indicates February 28, the day redemption was halted. Panel (b) reports coefficients from regressions of token acceptance on week fixed effects, with controls for user fixed effects and months after user entry. The sample includes only existing regular users. Standard errors are clustered at user level. 95% confidence intervals are shown.

finding is consistent with Prediction 3.1.

Token-mediated trades fell in the long run. Panel (a) in Figure 7 shows a slight decline in token-mediated trades in the weeks following the redemption halt. This gradual decline was interrupted by the arrival of the Covid pandemic, which led trades to fall sharply. As the lockdown eased over the next few months, token-mediated trade among existing regular users recovered most of the lockdown decline, but soon began to decline again. The long-run reduction among existing regular users is highly significant, even after controlling for individual fixed effects and time since entry. The aggregate series tells the same story: token-mediated trade reached roughly two-thirds of its pre-halt level in June 2020 and only one-third by the end of the year (Online Appendix Figure E1). These findings are consistent with Prediction 3.1.

Barter trade recovered after the lockdown. Unlike after the 2019 restriction, aggregate barter did not persistently fall. Figure 7 Panel (b) shows that barter among existing regular users fell during the lockdown and returned to close to its pre-halt level by the summer of 2020, consistent with the full recovery of aggregate barter in Online Appendix Figure E1. This matches Predic-

Figure 7: Effects of redemption halt on trade volumes, existing regular users



Notes: Figure reports coefficients from regressions of (a) asinh token-mediated trade and (b) asinh barter trade on week fixed effects, with controls for user fixed effects and months after user entry. The sample includes only existing regular users. Standard errors are clustered at user level. 95% confidence intervals are shown.

tions 3.2 and 3.3: with the reference already at zero, the halt added no aggrievement and moved participation only through the token's value, and that channel proved small.

Other measures of platform usage studied in the online appendix confirm the recovery. Offer messages sent substantially recovered, though not fully (Online Appendix Figure E2). The total number of new items posted among all users fully recovered, though some persistent decline is found among existing regular users, after controlling for user fixed effects and time since entry (Online Appendix Figure E3). Unlike in the 2019 episode, there was no discernible collapse in entry around the halt: the entry of regular users was already minimal in the months before the announcement and changed little thereafter (Online Appendix Figure B4).

7 Comparing the Two Crises

The previous two sections documented the platform's two reductions in redeemability. This section draws out what the comparison between the two events shows about trust, prices, and the adjustment of acceptance.

Table 3: The two redemption crises

	Restriction (Sept 2019)	Halt (Feb 2020)
Policy change	Redemption restricted to coffee and food merchants	Redemption ended entirely
Relation to reference point	Surprise breach: large fall in $\bar{p}$ relative to fixed $\bar{p}_0$	Reference already adjusted: fall to zero from an already-reduced level
Announcement and framing	Abrupt, reneging on a pledged 30-day notice; widely read as a betrayal	Framed as a temporary, technical pause; no comparable backlash
Token acceptance	Immediate fall, 35% $\rightarrow$ 27%	Immediate fall, 26% $\rightarrow$ 19%, then gradual decline
Token-mediated trade	Persistent decline, -32% within a month	Gradual decline to about one-third of the pre-halt level by late 2020
Barter trade	Persistent decline, -23% within a month; also within existing users	Fell during lockdown; aggregate recovered fully, existing users to near pre-halt level
User entry	Fell by roughly two-thirds	Already minimal before the halt; little further change
Posted prices (BTZ)	Median posted price at par	Median posted price at par

Notes: Estimates are those reported in Sections 5 and 6. The February 2020 long-run estimates are descriptive, since the episode coincided with the onset of the Covid-19 pandemic (Section 6).

7.1 Trust and Aggrievement

Table 3 sets the two episodes side by side. Both reduced token acceptance and token-mediated trade. They differed in the collateral damage: the restriction, which left redemption partly intact, reduced barter and entry persistently, whereas the halt, which ended redemption, did not. Why?

The theory points to aggrievement. Section 3 gives two channels through which a reduction in redeemability lowers barter: the token loses value, so the platform becomes less rewarding to use; and the broken promise aggrieves users, who withdraw. Canonical search models with fixed participation have only the first channel and predict, if anything, a rise in barter, since users impaired

in token trade substitute toward it. The 2020 halt removed the token’s remaining redemption value and yet left aggregate barter at its pre-halt level, barter among existing users close to it once the lockdown lifted, and entry unchanged. The token-value channel alone therefore cannot account for the persistence of the 2019 barter decline or the two-thirds fall in entry; the difference between the episodes is what aggrievement predicts (Predictions 2.3 and 3.3).

Qualitative evidence supports this reading. After the 2019 restriction, local media reported that users were angry the platform had reneged on its redemption commitment.²³ Some users publicly announced that they would depart from the platform, and many Bunz Facebook administrators announced that they would sever their affiliation with Bunz.²⁴ These accounts describe aggrieved users leaving the platform and tarnishing its reputation, the aggrievement channel of Prediction 2.2. No comparable backlash was observed after the 2020 halt, which drew none of the public protest or dissociation seen in 2019.

Three factors may explain why the halt did not aggrieve users. First, it was announced as a temporary, technical measure. As such, many users did not consider the reduction a breach of trust. Second, the users who most valued redemption may have already left after the 2019 restriction, thinning the population the halt could aggrieve. Third, users’ expectation that the remaining redemption would be honored may have already eroded before the breach.

7.2 Price Rigidity

A common intuition is that a reduction in redeemability should persistently reduce the token’s price. It did not in the Bunz setting. Posted token prices stayed within a few percent of the 100 BTZ to 1 CAD redemption rate throughout, and exactly at it after both redemption shocks, with many gift cards posted at exactly that rate despite dispersion from search frictions (Online Appendix

²³One article emphasized the disappointment of users, some of whom had been saving up their BTZ for bike repairs, records and other large purchases. For example, an administrator for several Bunz-related Facebook groups was quoted as saying that the announcement “felt like a punch in the gut.” She had amassed roughly \$600 worth of BTZ and treated the stockpile as a safety net in case she ran out of money and needed something for her two-year-old son. “Now, it’s worthless,” she said. “He doesn’t drink coffee” (Posadzki 2019).

²⁴As shown in Online Appendix D.4, this group renamed their Facebook groups as “PALZ”, writing, “Today, we would like to reclaim our communities. We would like to bring Bunz back to what it once was. We want our groups to remember why they exist. We do not want to profit. We do not want your app sign-ups. We do not want you to buy into an online currency that will let you down.”

Table 4: Equivalence test: no persistent shift in the median posted token price

	Coef.	90 percent CI	Eq. $\pm 10\%$	Eq. 80–125
Post-E1	−0.039	[−0.182, 0.025]	No	Yes
Post-E2	0.000	[−0.086, 0.039]	Yes	Yes
Post-E3	0.000	[−0.039, 0.039]	Yes	Yes

Notes: p is the posted BTZ price per CAD of gift-card face value; par is $p = 100$. Coef. is the change in the median of $\ln p$ after each event relative to the preceding regime, controlling for gift-card brand fixed effects, with its 90 percent month-cluster bootstrap confidence interval. E1, E2, and E3 denote the monetary expansion, the 2019 restriction, and the 2020 halt. “Eq.” equals Yes when the confidence interval lies entirely within the equivalence band $[\ln(0.90), \ln(1.10)]$ or $[\ln(0.80), \ln(1.25)]$. Online Appendix Table F1 reports the underlying estimates for the median, the quartiles, and the mean.

Figure F1).²⁵ A formal equivalence test rules out persistent median depegging larger than 10 percent following either redemption shock (Table 4). None of the estimated marginal effects of the three events on the median, the quartiles, or the mean of the posted price is statistically significant (Online Appendix Table F1). The adjustment fell instead on acceptance and quantities.

Why did posted prices not adjust? A potential explanation is that the price was held at par by arbitrage while the redemption program operated: a token redeemable for a fixed quantity of goods could not trade far from that value. But the price did not move even after redemption ended. A plausible resolution is that prices on the platform were posted in decentralized bilateral meetings with no centralized market, leaving no mechanism through which users could coordinate on a new level. Such indeterminacy is a standard property of decentralized monetary economies (Green and Zhou, 1998; Jean, Rabinovich and Wright, 2010; Kamiya and Shimizu, 2006) and recalls the classic indeterminacy of exchange rates between coexisting currencies (Kareken and Wallace, 1981). Here, the redemption peg supplied the anchor while it operated. A currency that trades in a centralized or secondary market can absorb a loss of backing through a fall in its price; a platform currency without one cannot, and the loss is expressed instead as a decline in acceptance.

²⁵Token prices are measured from gift cards carrying both a CAD face value and a BTZ price, specifically five frequently traded cards: Starbucks, Indigo, Apple iTunes, LCBO, and Amazon. Because BTZ was not freely exchangeable for cash, this is the best available proxy for the token price perceived by users. These are posted, not transaction, prices. One interviewed seller reports accepting BTZ at a 10 to 15 percent discount after the 2019 restriction (Section 5).

7.3 The Slow Adjustment

When redemption ended, acceptance of the token did not collapse to zero. It fell immediately, by 7 percentage points platform-wide, then eroded further over the following months, while the token continued to circulate at par despite carrying no redemption value. This persistence of unbacked circulation is a recognized phenomenon. It has been produced in laboratory economies, where a currency stripped of its redemption value remains in use (Duffy and Ochs, 2002), and observed historically in the Swiss dinar in Iraq (Foote et al., 2004; King, 2004) and the Somali shilling (Luther, 2015; Luther and White, 2016), which circulated for years after the authorities that had backed them were gone.

The pattern is consistent with two mechanisms from monetary theory. First, acceptance is self-fulfilling: a token is accepted partly because others accept it (Kiyotaki and Wright, 1992), so one that has lost its backing can still circulate as long as users expect others to keep accepting it, which is why acceptance did not fall to zero. Second, expectations adjust gradually rather than at once (Selgin, 2003), which is why acceptance eroded over time rather than dropping immediately to a new level.

8 Conclusion

This paper uses high-frequency transaction-level data to study trust and redeemability in a currency system, tracing the rise and fall of a redeemable platform currency through three unexpected monetary experiments. The five-fold supply expansion confirms that the token functioned as money, raising transaction volume persistently with no effect on barter. The two reductions in redeemability deliver three lessons. Redeemability sustained circulation: when the platform restricted and then ended redemption, token acceptance and token-mediated trade fell. Trust governed the collateral damage: the first breach, read by users as a betrayal, reduced barter and entry persistently, whereas the second, which elicited little protest, did not. The trust that makes a redemption promise credible is thus double-edged: it made circulation possible, but its loss amplified the damage. And the adjustment ran through acceptance rather than price: the token's median posted price stayed at par, and sellers refused the token rather than repricing it. Privately issued currencies without an

active secondary market may absorb redemption stresses the same way. A search-theoretic model of redeemable money with reference-dependent participation organizes the patterns.

References

- Alvarez, Fernando, David Argente and Diana Van Patten. 2023. “Are Cryptocurrencies Currencies? Bitcoin as Legal Tender in El Salvador.” *Science* 382(6677).
- Alvarez, Fernando E, David Argente, Francesco Lippi, Esteban Méndez and Diana Van Patten. 2023. Strategic Complementarities in a Dynamic Model of Technology Adoption: P2P Digital Payments. NBER Working Papers 31280 National Bureau of Economic Research, Inc.
- Alvarez, Fernando E. and David O. Argente. 2020*a*. Consumer Surplus of Alternative Payment Methods: Paying Uber with Cash. NBER Working Papers 28133 National Bureau of Economic Research.
- Alvarez, Fernando E. and David O. Argente. 2020*b*. On the Effects of the Availability of Means of Payments: The Case of Uber. NBER Working Papers 28145 National Bureau of Economic Research.
- Bernanke, Ben S. 2002. “Deflation: Making Sure “It” Doesn’t Happen Here.”. Speech before the National Economics Club, Washington, D.C.
- Borio, Claudio. 2019. On money, debt, trust and central banking. BIS Working Papers 763 Bank for International Settlements.
- Brunnermeier, Markus K. and Jonathan Payne. 2023. Strategic Money and Credit Ledgers. Working Paper 31561 National Bureau of Economic Research.
- Camera, Gabriele. 2024. “Introducing New Forms of Digital Money: Evidence from the Laboratory.” *Journal of Money, Credit and Banking* 56(1):153–184.
- Chiu, Jonathan and Tsz-Nga Wong. 2022. “Payments on digital platforms: Resiliency, interoperability and welfare.” *Journal of Economic Dynamics and Control* 142:104193.
- Clower, Robert. 1967. “A Reconsideration of the Microfoundations of Monetary Theory.” *Economic Inquiry* 6(1):1–8.
- Colacelli, Mariana and David J.H. Blackburn. 2009. “Secondary currency: An empirical analysis.” *Journal of Monetary Economics* 56(3):295–308.
- Cong, Lin William, Ye Li and Neng Wang. 2022. “Token-based platform finance.” *Journal of Financial Economics* 144(3):972–991.
- Duffy, John and Daniela Puzzello. 2014. “Gift Exchange versus Monetary Exchange: Theory and Evidence.” *American Economic Review* 104(6):1735–76.
- Duffy, John and Jack Ochs. 2002. “Intrinsically Worthless Objects as Media of Exchange: Experimental Evidence.” *International Economic Review* 43(3):637–674.
- Foote, Christopher, William Block, Keith Crane and Simon Gray. 2004. “Economic Policy and Prospects in Iraq.” *Journal of Economic Perspectives* 18(3):47–70.

- Friedman, Milton. 1969. *The optimum quantity of money, and other essays*. Chicago, Aldine Pub. Co.
- Galang, Jessica. 2019. “Bunz Locks Down Employee BTZ Wallets as Former Admins Lead Mass Exodus.” *BetaKit*. Accessed at <https://betakit.com/bunz-locks-down-employee-btz-wallets-as-former-admins-lead-mass-exodus/> on 2021-06-21.
- Gorton, Gary B. and Jeffrey Y. Zhang. 2023. “Taming Wildcat Stablecoins.” *University of Chicago Law Review* 90. Accessed at: <https://chicagounbound.uchicago.edu/uclrev/vol90/iss3/3>.
- Graeber, David. 2011. *Debt: The First 5000 Years*. Brooklyn, N.Y. : Melville House.
- Green, Edward J. and Ruilin Zhou. 1998. “A Rudimentary Random-Matching Model with Divisible Money and Prices.” *Journal of Economic Theory* 81(2):252–271.
- Gu, Chao, Fabrizio Mattesini, Cyril Monnet and Randall Wright. 2013a. “Banking: A New Monetarist Approach.” *The Review of Economic Studies* 80(2):636–662.
- Gu, Chao, Fabrizio Mattesini, Cyril Monnet and Randall Wright. 2013b. “Endogenous Credit Cycles.” *Journal of Political Economy* 121(5):940–965.
- Hart, Oliver and John Moore. 2008. “Contracts as Reference Points.” *The Quarterly Journal of Economics* 123(1):1–48.
- Humphrey, Caroline. 1985. “Barter and Economic Disintegration.” *Man* 20(1):48–72.
- Jack, William and Tavneet Suri. 2014. “Risk Sharing and Transactions Costs: Evidence from Kenya’s Mobile Money Revolution.” *American Economic Review* 104(1):183–223.
- Jean, Kasie, Stanislav Rabinovich and Randall Wright. 2010. “On the multiplicity of monetary equilibria: Green–Zhou meets Lagos–Wright.” *Journal of Economic Theory* 145(1):392–401.
- Jiang, Janet Hua, Peter Norman, Daniela Puzzello, Bruno Sultanum and Randall Wright. 2024. “Is Money Essential? An Experimental Approach.” *Journal of Political Economy* 132(9):2972–2998.
- Jing, Baiyun, Michael B. Wong, Yang You and Yulin Zhong. 2026. “Redemption Design and Digital Currency Adoption.” Working paper.
- Kamiya, Kazuya and Takashi Shimizu. 2006. “Real indeterminacy of stationary equilibria in matching models with divisible money.” *Journal of Mathematical Economics* 42(4-5):594–617.
- Kareken, John and Neil Wallace. 1981. “On the Indeterminacy of Equilibrium Exchange Rates.” *Quarterly Journal of Economics* 96(2):207–222.
- Kehoe, Timothy J. and David K. Levine. 1993. “Debt-Constrained Asset Markets.” *The Review of Economic Studies* 60(4):865–888.

- Keister, Todd and Daniel Sanches. 2023. "Should Central Banks Issue Digital Currency?" *The Review of Economic Studies* 90(1):404–431.
- King, Mervyn. 2004. "The Institutions of Monetary Policy." *American Economic Review* 94(2):1–13.
- Kiyotaki, Nobuhiro and Randall Wright. 1989. "On Money as a Medium of Exchange." *Journal of Political Economy* 97(4):927–954.
- Kiyotaki, Nobuhiro and Randall Wright. 1992. "Acceptability, means of payment, and media of exchange." *Quarterly Review* .
- Kiyotaki, Nobuhiro and Randall Wright. 1993. "A Search-Theoretic Approach to Monetary Economics." *The American Economic Review* 83(1):63–77.
- Lagos, Ricardo, Guillaume Rocheteau and Randall Wright. 2017. "Liquidity: A New Monetarist Perspective." *Journal of Economic Literature* 55(2):371–440.
- Lagos, Ricardo and Randall Wright. 2005. "A Unified Framework for Monetary Theory and Policy Analysis." *Journal of Political Economy* 113(3):463–484.
- Liu, Jiageng, Igor Makarov and Antoinette Schoar. 2023. Anatomy of a Run: The Terra Luna Crash. NBER Working Papers 31160 National Bureau of Economic Research.
- Luther, William J. 2015. "The monetary mechanism of stateless Somalia." *Public Choice* 165(1/2):45–58.
- Luther, William J. and Lawrence H. White. 2016. "Positively Valued Fiat Money after the Sovereign Disappears: The Case of Somalia." *Review of Behavioral Economics* 3(3-4):311–334.
- Nielsen, Kevin. 2020. "A timeline of COVID-19 in Ontario." *Global News* . Accessed at <https://globalnews.ca/news/6859636/ontario-coronavirus-timeline/> on 2021-06-21.
- Ozdenoren, Emre, Yuan Tian and Kathy Yuan. 2025. "Platform Money." Working Paper, last revised: May 9, 2025.
- Posadzki, Alexandra. 2019. "'It felt like a punch in the gut': Scale-back of cryptocurrency venture Bunz stuns businesses, clients." *Globe and Mail* . Accessed at <https://www.theglobeandmail.com/canada/toronto/article-it-felt-like-a-punch-in-the-gut-scale-back-of-cryptocurrency/> on 2021-06-21.
- Radford, Robert A. 1945. "The Economic Organisation of a P.O.W. Camp." *Economica* 12(48):189–201.
- Rogoff, Kenneth and Yang You. 2023. "Redeemable Platform Currencies." *The Review of Economic Studies* 90(2):975–1008.

- Rupert, Peter, Martin Schindler, Andrei Shevchenko and Randall Wright. 2000. "The search-theoretic approach to monetary economics: a primer." *Federal Reserve Bank of Cleveland Economic Review* 36(Q IV):10–28.
- Schnabel, Isabel and Hyun Song Shin. 2018. Money and trust: lessons from the 1620s for money in the digital age. BIS Working Papers 698 Bank for International Settlements.
- Selgin, George. 2003. "Adaptive Learning and the Transition to Fiat Money." *Economic Journal* 113(484):147–165.
- Shevchenko, Andrei and Randall Wright. 2004. "A simple search model of money with heterogeneous agents and partial acceptability." *Economic Theory* 24(4):877–885.
- Sockin, Michael and Wei Xiong. 2023. "Decentralization through Tokenization." *The Journal of Finance* 78(1):247–299.
- Sweeney, Joan and Richard James Sweeney. 1977. "Monetary Theory and the Great Capitol Hill Baby Sitting Co-op Crisis: Comment." *Journal of Money, Credit and Banking* 9(1):86–89.
- TTC Board. 2020. "TTC's Response to COVID-19." TTC Board meeting materials. Accessed at <https://www.ttc.ca/All-public-meetings/board/2020/Board-Meetings-May-13?tab=0> on 2021-06-21.
- Wray, L. Randall, ed. 2004. *Credit and State Theories of Money*. Edward Elgar Publishing.
- Wright, Randall. 1999. "A note on asymmetric and mixed strategy equilibria in the search-theoretic model of fiat money." *Economic Theory* 14(2):463–471.

Appendix

A Model Details

A.1 Value Functions

Each agent either holds one unit of money or none. Agents choose participation a and whether to accept commodities and money in order to maximize expected discounted utility, taking as given the strategies of others. In a symmetric stationary equilibrium in which all agents choose participation a and the platform survives with probability s , all redemption-related terms scale by the effective redeemability $\tilde{\rho} = s\rho$: the model with failure risk is isomorphic to the baseline with ρ replaced by $\tilde{\rho}$. The Bellman equations are:

$$rV_1 = \alpha a^2 xy(u - c) + \alpha a^2 x(1 - y)(1 - M)\pi(u + V_0 - V_1) + \tilde{\rho}(u + V_0 - V_1) - \Gamma(a) \quad (8)$$

$$rV_0 = \alpha a^2 xy(u - c) + \alpha a^2 x(1 - y)\pi M(V_1 - V_0 - c) + \sigma(V_1 - V_0) - \Gamma(a) \quad (9)$$

where V_1 is the value of holding one unit of money, V_0 the value of holding none, and $\Gamma(a) \equiv C(a) + \lambda aG$ is the total flow cost of participation (effort plus aggrievement). The first term in each equation is the utility flow from barter, the second from monetized exchange, and the third from money redemption or issuance. Setting $a = 1$, $s = 1$, and $\lambda = 0$ recovers the benchmark model without the participation, failure-risk, and aggrievement extensions.

The variables are defined as follows: $u > c$ is consumption utility, $c > 0$ is production cost, $x \in (0, 1)$ is the probability that an agent is able to consume the other agent's product, and xy is the probability that the pair has "double coincidence of wants," with $y \in (0, 1)$. $M \in [0, 1]$ is the fraction of agents endowed with money. $\tilde{\rho}$ denotes the effective rate of arrival of usable redemption opportunities. The rate of money issuance to agents without money is $\sigma = \frac{M}{1-M}\tilde{\rho}$, which holds the money supply constant, roughly matching the empirical setting, where token supply was largely stable except for two short periods of monetary expansion. Agents discount at rate $r > 0$.

Because participation cannot be conditioned on money holdings, the barter flow and the cost

$\Gamma(a)$ are common to both states and cancel from $V_1 - V_0$; the acceptance problem is therefore identical to the baseline model with α replaced by αa^2 and ρ by $\tilde{\rho}$.

To rule out the unrealistic possibility that agents without money prefer to wait for money issuance rather than accept money, we assume that $M < \bar{M} \equiv 1 - c/u$. This possibility arises because agents can accumulate at most one unit of money, so only those without money can receive a helicopter drop. This opportunity cost is especially large when a large share of agents hold money, so M must be bounded from above.

A.2 Proof of Proposition 1

Denote π_0 as the probability of accepting money in exchange for a commodity and π_1 as the probability of paying money for a commodity, so $\pi = \pi_0 \pi_1$. Let $\Delta_0 = V_1 - V_0 - c$ and $\Delta_1 = u + V_0 - V_1$. It follows that:

$$\pi_j = \begin{cases} 1 \\ \in [0, 1] \\ 0 \end{cases} \iff \Delta_j \begin{cases} > 0 \\ = 0 \\ < 0, \end{cases} \quad (10)$$

where

$$\Delta_0 = \frac{\alpha a^2 x(1-y)(1-M)\pi(u-c) + \tilde{\rho}(u-c) - (r+\sigma)c}{r + \tilde{\rho} + \sigma + \alpha a^2 x(1-y)\pi}, \quad (11)$$

$$\Delta_1 = \frac{(r+\sigma + \alpha a^2 x(1-y)M\pi)(u-c) + (r+\sigma)c}{r + \tilde{\rho} + \sigma + \alpha a^2 x(1-y)\pi}. \quad (12)$$

It is always the case that $\Delta_1 > 0$. Let

$$\bar{\rho} = \frac{rc(1-M)}{u(1-M)-c}, \quad \underline{\rho} = \frac{rc(1-M) - \alpha a^2 x(1-y)(1-M)^2(u-c)}{u(1-M)-c}, \quad (13)$$

where a is the participation level of the monetary equilibrium, $a^*(\pi = 1)$. Both $\bar{\rho} > 0$ and $\bar{\rho} > \underline{\rho}$ follow from $u(1-M) > c$ and $a > 0$. If $\tilde{\rho} > \bar{\rho}$, then $\Delta_0 > 0$ regardless of π . If $\tilde{\rho} < \underline{\rho}$, then $\Delta_0 < 0$ regardless of π . Note that $\bar{\rho}$ does not depend on participation: the threshold above which

only monetary equilibria exist is invariant to a , while $\underline{\rho}$ rises (the coexistence region narrows) as participation falls.

A.3 Mixed-Strategy Equilibria

Lemma A.1 (Mixed acceptance). *For $\tilde{\rho} \in (\underline{\rho}, \bar{\rho})$, there is exactly one stationary equilibrium with interior acceptance, given by $\pi_1 = 1$ and*

$$\pi_0^* = \frac{rc(1-M) - \tilde{\rho}[u(1-M) - c]}{\alpha a^2 x(1-y)(1-M)^2(u-c)}. \quad (14)$$

It has the following properties: (i) π_0^ is strictly decreasing in $\tilde{\rho}$, with $\pi_0^* \rightarrow 1$ as $\tilde{\rho} \rightarrow \underline{\rho}$ and $\pi_0^* \rightarrow 0$ as $\tilde{\rho} \rightarrow \bar{\rho}$; (ii) the equilibrium is unstable: under any monotone adjustment dynamic, acceptance slightly above π_0^* converges to the monetary equilibrium and acceptance slightly below converges to the non-monetary equilibrium.*

Proof. Since $\Delta_1 > 0$ always, $\pi_1 = 1$, and an interior equilibrium requires $\Delta_0 = 0$. Setting the numerator of (11) to zero and substituting $\sigma = \frac{M}{1-M}\tilde{\rho}$ yields (14); the stated limits follow from (13), and $\partial\pi_0^*/\partial\tilde{\rho} = -[u(1-M) - c]/[\alpha a^2 x(1-y)(1-M)^2(u-c)] < 0$ since $M < \bar{M}$. For (ii), note that at $\pi = \pi_0^*$ the numerator of (11) is zero and is strictly increasing in π , so $\Delta_0 > 0$ (best response: accept) for $\pi > \pi_0^*$ and $\Delta_0 < 0$ (best response: reject) for $\pi < \pi_0^*$. $\square$

Property (i) confirms that the comparative statics of interior acceptance are counter-intuitive, since as redeemability falls acceptance must *rise* to keep the marginal seller indifferent, but property (ii) implies that the interior equilibrium is the tipping point between the basins of attraction of the two pure equilibria rather than a state in which the economy can persist. Persistent partial acceptance instead arises naturally from heterogeneity (Section A.5 below).

A.4 Participation

Consider an agent choosing participation a_i when all others choose $\bar{a}$. Meetings arrive at rate $\alpha a_i \bar{a}$. The expected surplus per meeting is $xy(u-c)$ from barter in either money state; in single-coincidence meetings, a money holder earns Δ_1 when meeting an accepting non-holder (probability $x(1-y)(1-M)\pi$) and a non-holder earns $\Delta_0 + c - c = \Delta_0$ when accepting from a holder (probability $x(1-y)M\pi$). Averaging over the stationary distribution of money holdings ($M, 1-M$) and

using $\Delta_0 + \Delta_1 = u - c$:

$$\bar{S}(\pi) = xy(u - c) + x(1 - y)M(1 - M)\pi(u - c), \quad (15)$$

as in equation (1) of the main text. The agent's flow objective in a_i is $\alpha a_i \bar{a} \bar{S} - C(a_i) - \lambda a_i G$, with first-order condition $\alpha \bar{a} \bar{S} = C'(a_i) + \lambda G$. Imposing symmetry $a_i = \bar{a} = a^*$ gives equation (2). The comparative static (3) follows from the implicit function theorem at the selected solution, where generically $C''(a^*) > \alpha \bar{S}$; $a = 0$ is also an equilibrium, since no one participates when no one else does, and is not selected.

Trade flows are $T_B = \frac{1}{2} \alpha a^{*2} xy$ and $T_M = \alpha a^{*2} x(1 - y)(1 - M)M\pi$. Prediction 1 follows from $\tilde{\rho} > \bar{\rho}$ at the post-expansion stock (so $\pi = 1$ before and after) and $M < \min\{\bar{M}, \frac{1}{2}\}$ before and after: $\partial T_M / \partial M > 0$ directly, and a^* rises weakly since $\partial \bar{S} / \partial M = x(1 - y)(1 - 2M)\pi(u - c) > 0$. Prediction 2 follows from an unanticipated fall in $\tilde{\rho}$ into or below the coexistence region, with $\tilde{\rho}' < \tilde{\rho}_0$: acceptance falls (in the benchmark because the monetary equilibrium is eliminated or its tipping point rises, and in the extension of Section A.5 monotonically), $\bar{S}$ falls, and both terms of (3) reduce a^* ; because grievance persists once incurred, the lower level persists. Prediction 3 follows from setting $\tilde{\rho}_0 = 0$, since the halt is anticipated, and lowering $\tilde{\rho}$ to zero: G does not rise, so a^* responds through the token-value term alone, which removes the remaining monetary component of $\bar{S}$.

A.5 Heterogeneous Redemption Access

This section develops the extension referenced in Section 3, in which persistent partial acceptance arises from heterogeneity in redemption access, in the spirit of [Shevchenko and Wright \(2004\)](#). A richer treatment of heterogeneity, including money flows across agents and redemption policy design, is developed in a companion paper ([Jing et al., 2026](#)).

Setup. Agents differ in their effective redemption rate $\tilde{\rho}_i = s\rho_i$, where ρ_i reflects, for example, proximity to token-redeeming merchants. Let ρ_i be distributed across agents with c.d.f. F , fixed over time for each agent. To preserve tractability, each type's money-holder share is held at M :

redeemed tokens are reissued within type at rate σ_i , and the net flow of tokens across types through trade is neglected.²⁶ Let $\Pi \equiv \int \pi_0^i dF(i)$ denote aggregate acceptance among non-holders, which is what a money holder cares about when matching.

Individual acceptance. Let $D_i \equiv V_1^i - V_0^i$. Subtracting the type- i Bellman equations (with the aggregate Π replacing π on the spending side and own π_0^i on the acceptance side):

$$rD_i = [\alpha a^2 x(1-y)(1-M)\Pi + \tilde{\rho}_i] (u - D_i) - \alpha a^2 x(1-y)M\pi_0^i(D_i - c) - \sigma_i D_i, \quad (16)$$

with $\sigma_i = \frac{M}{1-M}\tilde{\rho}_i$. A type accepts ($\pi_0^i = 1$) if $D_i > c$ and rejects if $D_i < c$. For D_i in the neighborhood of c , D_i is strictly increasing in $\tilde{\rho}_i$ (this uses $M < \bar{M}$) and strictly increasing in Π , so acceptance has a threshold structure with strategic complementarity. Setting $D_i = c$ (at which point the own-acceptance term vanishes) and solving:

$$\tilde{\rho}^\dagger(\Pi) = \frac{(1-M)[rc - \alpha a^2 x(1-y)(1-M)\Pi(u-c)]}{u(1-M) - c} = \bar{\rho} - (\bar{\rho} - \underline{\rho})\Pi, \quad (17)$$

a linear, decreasing function of aggregate acceptance, with $\tilde{\rho}^\dagger(0) = \bar{\rho}$ and $\tilde{\rho}^\dagger(1) = \underline{\rho}$: the heterogeneous model nests the thresholds of Proposition 1, and the pure-strategy benchmark is the degenerate- F limit.

Equilibrium. Type i accepts iff $s\rho_i \geq \tilde{\rho}^\dagger(\Pi)$, so stationary equilibria are fixed points of

$$\Pi = \Phi_s(\Pi) \equiv 1 - F(\tilde{\rho}^\dagger(\Pi)/s), \quad (18)$$

with Φ_s increasing in Π (complementarity) and in s .

Proposition A.1 (Partial acceptance under heterogeneity). *In the heterogeneous-access economy, holding each type's money-holder share fixed at M :*

1. (Cross section.) *In any equilibrium, acceptance is monotone in redemption access: $\pi_0^i = \mathbf{1}\{\rho_i \geq \tilde{\rho}^\dagger(\Pi)/s\}$.*

²⁶Accounting for that flow, money accumulates among accepting types and the stationary distribution of holdings varies by type; the threshold structure below survives at the cost of additional notation, as in the discrete-time treatment of [Jing et al. \(2026\)](#).

2. (Stable partial acceptance.) *An interior equilibrium $\Pi^* \in (0, 1)$ is stable under best-response dynamics iff $f(\tilde{\rho}^\dagger(\Pi^*)/s) \cdot (\bar{\rho} - \underline{\rho})/s < 1$: sufficient dispersion of redemption access delivers stable interior acceptance. Multiple equilibria (including coexistence of high- and low-acceptance regimes) arise when F concentrates mass inside the coexistence region.*
3. (Smooth decline, discontinuous collapse.) *At any stable interior equilibrium, a marginal fall in s (or a uniform downward shift of redemption access) reduces Π^* continuously and monotonically. A large fall in s can eliminate the high-acceptance equilibrium altogether (a tangency of Φ_s with the diagonal), producing a discontinuous collapse onto a lower equilibrium.*

Proof. (1) is the threshold characterization above. (2): Φ_s maps $[0, 1]$ into itself and is increasing, so fixed points exist; local stability of $\Pi_{t+1} = \Phi_s(\Pi_t)$ requires $\Phi'_s(\Pi^*) = f(\tilde{\rho}^\dagger/s) \cdot (\bar{\rho} - \underline{\rho})/s < 1$, using (17). (3): at a stable fixed point, $d\Pi^*/ds = \frac{\partial\Phi_s/\partial s}{1-\Phi'_s} > 0$ and finite, by the implicit function theorem; when the fixed point becomes tangent ($\Phi'_s \rightarrow 1$), it disappears and the dynamics converge to the next-lowest fixed point. $\square$

Proposition A.1 delivers the features of the data that the homogeneous benchmark cannot: acceptance that declines partially, but does not collapse, after a moderate deterioration in redeemability (part 3, small shift), the possibility of run-like discontinuous collapse after a large one (part 3, tangency), and persistent interior acceptance throughout (part 2). It also implies that acceptance should be higher among users with better redemption access, a cross-sectional prediction examined in Jing et al. (2026).